

Topological Quantum Neural Networks and ZX-Calculus: From Spin-Networks to Categorical Case Diagrams

Topological Quantum Neural Networks and ZX-Calculus: From Spin-Networks to Categorical Case Diagrams I

Where we are in the argument. The preceding sections have established that categorical string diagrams—from DisCoCat's pregroup derivations (??) through enriched hom-values (??) to topos-theoretic transfer (??)—provide a unified diagrammatic language for case-theoretic reasoning (**theoretical synthesis**). This section extends the framework with a literature bridge to topological quantum neural networks (TQNNs), the ZX-calculus, and sheaf-theoretic quantum semantic communication (**theoretical bridge to prior art**).

Topological Quantum Neural Networks and ZX-Calculus: From Spin-Networks to Categorical Case Diagrams II

The `src/quantum/quantum_case.py` module implements a concrete POVM-based measurement model for case roles (tested in the project suite; see ??). The broader TQNN, ZX-rewrite, and lambeq compilation claims remain literature connections and proposed extensions rather than full local implementations. The central observation is that the same monoidal-categorical architecture underlies both classical DisCoCat diagrams and quantum processes.

Topological Quantum Neural Networks and ZX-Calculus: From Spin-Networks to Categorical Case Diagrams III

Position relative to prior work. Each ingredient we draw on has independent prior art. TQNNs as spin-network computations are due to Fields, Marcianò, and collaborators [fields2022tqnn; fields2025amplituhedra]; the ZX-calculus and its circuit-extraction theory are due to Kissinger, van de Wetering, Coecke, and the Picturing-Quantum-Processes school [kissinger2020zx; coecke2017]; and quantum natural language processing on present-day hardware has been demonstrated by the lambeq pipeline [lorenz2021lambeq] and the Quantinuum quantum-NLP programme. What is *new* in this section is none of those constructions individually; rather it is the **identification** that the *same* monoidal-functorial scaffolding can carry a *case-theoretic* payload — in particular, that the POVM family $\{E_c\}$ of ?? can be read both as a Born-rule case-assignment device and as the pointer-basis selected by a quantum reference frame in a TQNN — and the consequent claim that case assignment, ZX-rewrite verification, and DAIF-style distributional

QNNs as Spin-Networks I

QNNs as Spin-Networks II

TQFT as the Forward Pass: Reshetikhin–Turaev Invariants Compute Network Amplitudes

Marcianò, Fields, and Glazebrook show that quantum neural networks (QNNs) admit a topological reformulation using spin-networks [fields2022tqnn]. Any QNN layer can be represented as a graph whose edges carry representation labels (spins) and whose vertices carry intertwiners—precisely the data defining a spin-network in a 3-dimensional topological quantum field theory (TQFT):

“Quantum Neural Networks (QNNs) can be mapped onto spin-networks, with the consequence that the level of analysis of their operation can be carried out on the side of Topological Quantum Field Theory (TQFT).” [fields2022tqnn]

This reformulation has three structural consequences. First, the network becomes a topological diagram—spin-network or ribbon graph—evaluated by a continuous TQFT functor; edges encode information flow and nodes encode transformation. Second, the TQFT evaluation assigns boundary Hilbert spaces to the diagram via the Reshetikhin–Turaev and Turaev–Viro invariants, playing the

ZX-Calculus: Topological String Diagrams Where Graph Rewrites Are Quantum Proofs I

ZX-Calculus: Topological String Diagrams Where Graph Rewrites Are Quantum Proofs II

String Diagrams for Quantum Processes

The ZX-calculus provides a diagrammatic language for quantum circuits, representing them as string diagrams in a symmetric monoidal category of finite-dimensional Hilbert spaces and linear maps [Kissinger2020zx]:

“The ZX-calculus is a graphical language for reasoning about quantum computations and circuits... it can represent any linear map, and can be considered a diagrammatically complete generalization of the usual circuit representation.” [CoeckeKissinger2017]

Three structural features connect ZX to case-theoretic diagrams:

- ▶ **Diagrams as morphisms:** ZX-diagrams are string diagrams in a $\dagger$ -compact closed category. Wires represent objects (qubits), spiders and boxes represent morphisms, and composition/tensor product correspond to vertical/horizontal concatenation. Only topology matters, not geometry.
- ▶ **Deterministic circuit extraction via generalized flow:**

One Diagram, Three Interpretations: TQNN, ZX, and DisCoCat Share a Monoidal Functor I

One Diagram, Three Interpretations: TQNN, ZX, and DisCoCat Share a Monoidal Functor II

A Common Language of Ribbon and Tensor Diagrams

Both ZX-diagrams and TQNNs are topological string diagrams evaluated by monoidal functors into the category of Hilbert spaces and linear maps. The alignment becomes explicit when stated precisely:

- ▶ **For TQNNs:** A 3-dimensional TQFT functor from a cobordism or skein category to **Hilb** assigns to each spin-network/ribbon graph a linear map representing the TQNN computation. The underlying topological skein theory treats network layers as *ribbon graphs* whose evaluation via Reshetikhin–Turaev and Turaev–Viro invariants gives quantum processes implementing computation and error correction [fields2022tqnn; fields2025amplituhedra].
- ▶ **For ZX:** The standard semantics functor from the free $\dagger$ -compact category generated by Z/X spiders, Hadamards, etc. into **FHilb** assigns each ZX-diagram a linear map [kissinger2020zx].